

$$\sigma = \frac{1}{a} \sqrt{\frac{\beta^2 \tau c F^0}{k_{10} + \beta}}$$

$$\frac{d\sigma}{dt} = \frac{\partial \sigma}{\partial a} \frac{da}{dt} + \frac{\partial \sigma}{\partial \beta} \frac{d\beta}{dt}$$

$$\frac{d\sigma}{dt} = -\frac{\beta}{a^2} \sqrt{\frac{\tau c F^0}{k_{10} + \beta}} \frac{da}{dt} + \frac{\sqrt{\tau c F^0} (2k_{10} + \beta)}{2a(k_{10} + \beta)^{3/2}} \frac{d\beta}{dt}$$